Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
B		2600308

Moon Trek: A Hybrid Logistics System for Moon Colony

Summary

As humanity moves toward sustained extraterrestrial habitation, constructing a **Moon Colony** has become a large-scale optimization problem. The central challenge is transporting massive resources and long-term supplies to the Moon under coupled cost, time, and environmental constraints. To address this, we build a multi-dimensional framework that evaluates alternative delivery scenarios, identifies robustness and supply schemes under realistic limitations.

For Task I, we propose a **multi-dimensional optimization** model to improve delivery methods, balancing the transportation cost and time. Considering the realistic parameters in the aerospace industry and the progress of technology, we integrate the baseline of rocket-only and space elevator-only into a **hybrid logistics framework**. Compared to the baseline, the efficiency rose about **37.8%** and **60.5%** respectively, at $\lambda = 0.617$. Hence, a **trade-off balance** statement is constituted under two constraints.

For Task II, we consider more realistic conditions, transferring the model to non-ideal cases. Using **probability analysis**, potential risks are integrated into the proposed model, providing detailed strategy adjustment. In addition, **Monte Carlo simulation** is applied to check the feasibility. A **Policy Switching Index** of **2.5%** indicates that the optimal strategy remains stable under moderate variations in decision-maker priorities, demonstrating strong robustness.

For Task III, the problem is formulated into a **Markov decision process** to determine how to keep the water consumption of 100,000 people in the Moon Colony. With multiple constraints, we select a reinforcement learning method, which can solve the problem of **long-term decision-making**, promising the water supply. Moreover, a **water circulation system** is combined, making it more consistent with realistic technological capabilities.

For Task IV, we decompose environmental impact into multiple components and integrate them into a **unified framework**. This function is then embedded into the hybrid logistics framework to construct an **environment-augmented model**. We further introduce the **Environmental Efficiency Index** to evaluate environmental cost under an optimal strategy. The hybrid model achieves the lowest EEI **($\approx$ 0.22)** compared to the baseline configuration. It represents a minimal environmental cost while securing the environmental protection effect.

Finally, we evaluate the performance and advantage of the proposed models, using **sensitivity and robustness analysis**. Though the methods do not consider all the factors, it still enlightens the study to construct a moon colony. Based on these analyses, we present available recommendations in the policy-oriented letter, providing practical guidance for the governments to create the Moon Colony.

Keywords: Moon Colony; Multi-dimensional optimization; Monte Carlo; Markov decision process; Environment-augmented model

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1 Introduction

1.1 Problem Background

We choose to go to the moon, not because they are easy, but because they are hard
— John F. Kennedy, 1962

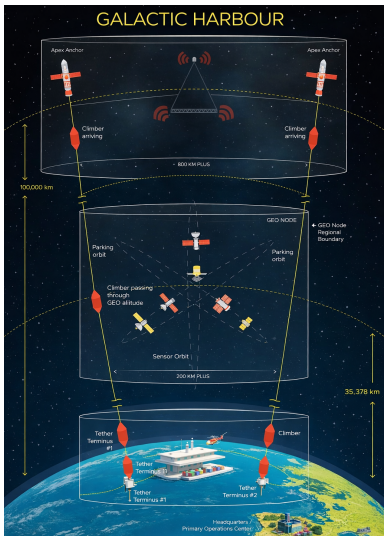
Humanity's pursuit of space exploration has gradually shifted from short-term scientific missions toward large-scale, sustained extraterrestrial habitation[1, 2]. Among these ambitions, a permanent Moon Colony capable of supporting a population of around 100,000 represents a meaningful milestone. Such a settlement is envisioned not only as a research station, but also as a long extension of human civilization. It alleviates pressure on Earth's fragile ecosystems while enabling industrial and commercial activities beyond the planet [3].

Under conventional rocket-launch paradigms, the requirements of the Moon Colony undoubtedly translate into enormous economic costs. To overcome the challenges, the concept of a Space Elevator has emerged as an alternative to traditional methods, as shown in Figure 1. From there, resources can be transferred to the Moon using tremendously reduced rocket propellant compared to direct Earth-to-Moon launches.

Despite its benefits, the Space Elevator introduces a new set of strategic and operational questions. There are uncertainties such as elevator breakdowns, tether sway, and environmental impact. Consequently, decision-makers encounter the trade-offs of whether to rely exclusively on space elevators, traditional rockets, or a mixed strategy to meet the Moon Colony's construction and sustainment demands.



(a) Space Elevator Concept Art



(b) space elevator diagram

Figure 1: Space Elevator

1.2 Restatement of the Problem

Considering the background information and restricted conditions identified in the problem statement, we need to solve the following problems:

- **Task I: Compare delivery scenarios for construction materials**
  Build a model to quantify the total cost and time required under three delivery scenarios, ranging from using a space elevator only, traditional rocket launches only, to the combination of both methods

- **Task II: Evaluate performance under non-ideal conditions**
  Extend the model to non-ideal conditions, and quantify how the reliability of cost changes with respect to each method.
- **Task III: Estimate one-year water supply sustainability**
  Determine the annual amount of water required to sustain a Moon Colony with around $100,000$-person, and find out the extra cost and timeline for water supply.
- **Task IV: Assess environmental impacts and propose solutions**
  Evaluate the impact on Earth's environment with respect to the delivery mission and adjust the previous model to reduce the environmental impact.

1.3 Our Work

This problem requires us to construct a model to specify the delivery strategies, both economical and environmentally friendly. Our work can be illustrated with the following flowchart in Figure 2.

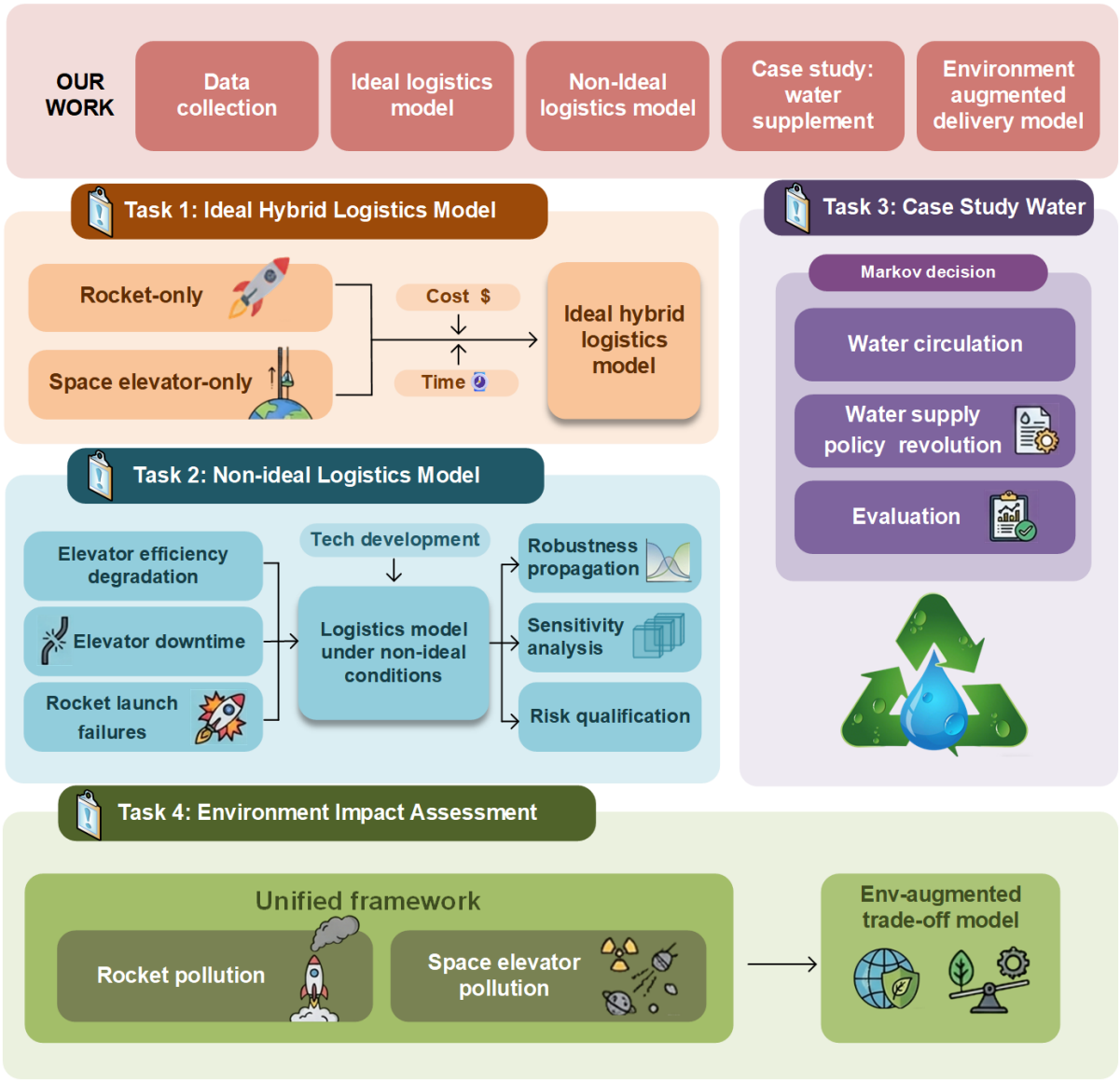


Figure 2: Flow chart of our work

2 Assumptions and Justifications

To simplify the problems, we make the following basic assumptions with proper justification.

- **Assumption:** The angle between the equatorial plane and the ecliptic plane is fixed at its mean value of $23.26°$.
  Justification: Earth's axial tilt variation is negligible over the mission timescale, so a constant-angle approximation is sufficient for orbital analysis.
- **Assumption:** All rocket launches occur at daily optimal launch windows that minimize orbital transfer energy.
  Justification: This reflects standard trajectory optimization practice and captures the theoretical minimum-energy transfer.
- **Assumption:** The single-launch payload capacity is fixed at 125 metric tons.
  Justification: This value represents the expectation of the specified 100–150 metric ton range, providing a stable and unbiased baseline.
- **Assumption:** All selected launch sites operate at their maximum feasible launch rate, with up to two launches per day under ideal conditions.
  Justification: This upper-bound assumption isolates transportation capacity as the primary system constraint rather than scheduling limitations.

Additional assumptions are made to simplify the analysis for individual sections. These assumptions will be discussed at the appropriate locations.

3 Notations

The symbols in Table 1 are frequently used in the model construction. Others will be introduced when they are used.

Table 1: Notations used in this paper

Symbol	Description	Units
v	velocity of the rocket	m/s
m	the mass of the rocket	kg
r_E	the radius of the Earth	m
R_M	the rotation radius of the Moon	m
ϕ	the latitude angle of the rocket launch Sites	degree
c	the capacity of the rocket	metric tons/year
C	the cost of the mission	USD/ton
m_0	the total mass of the rocket including propellant	kg
m_1	the mass of the Payload	kg

4 Data Collection

The quantitative parameters in this study are obtained from publicly available sources, including international statistics, space agency technical reports, and industrial disclo-

sures, and are used for cost, capacity, and energy modeling. The main data sources are summarized in Table 2.

Table 2: Data and Database Websites

Database Names	Database Websites
Rocket Launch Cost and Payload Capacity (NASA)	https://www.nasa.gov/
Commercial Rocket Cost (SpaceX)	https://www.spacex.com/
Space Elevator Technical and Feasibility Studies	https://www.isec.org/
Industrial Electricity Price Data	https://www.iea.org/
Fuel Price Reference Data	https://www.eia.gov/

5 Task I: A Cislunar logistics system

This section develops a unified logistics model to quantify the total cost and completion time of transporting construction materials from the Earth to the Moon [6].

- Space elevator-only delivery model
- Rocket-only delivery model

These two scenarios serve as a baseline and provide the foundation for a subsequent hybrid delivery model that combines both methods [4, 5].

- Hybrid logistics model

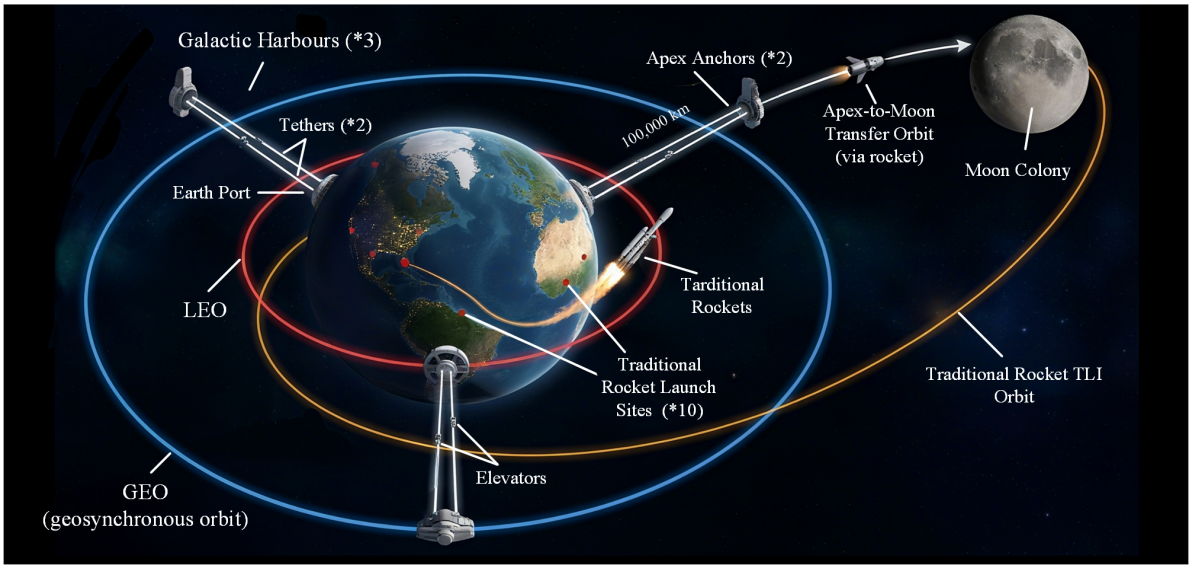


Figure 3: The baseline of two scenarios

5.1 Baseline Delivery Scenarios

5.1.1 Space Elevator–Only Delivery Model

In the space elevator–only scenario, all construction materials are delivered through the Space Elevator System, with no Earth-based rocket launches. That is, the total mass is undertaken by the elevator, where m_r and m_e represent the total mass delivered by

rockets and elevators, respectively.

$$m_e = D_{\text{total}} = 10^8 \text{ tons}, \qquad m_r = 0$$

The Space Elevator System consists of three Galactic Harbours, each with an annual lifting capacity of $179{,}000$ metric tons. Hence, the total lifting capacity of the system is

$$c_E = 3 \times 179{,}000 \ = 537{,}000 \text{tons/year} \tag{1}$$

Assuming continuous and uninterrupted operation, the total delivery time required to transport all materials from Earth to the apex anchors is

$$T_e = \frac{m_e}{c_E} = \frac{D_{\text{total}}}{c_E} \tag{2}$$

We first consider the cost related to the elevator lifting process. Let c_e represent the unit cost per ton transported by the space elevator. Since the elevator operates with a linear cost structure, the total elevator lifting cost is shown as:

$$C_{e,\text{lift}} = C_e m_e = C_e D_{\text{total}} \tag{3}$$

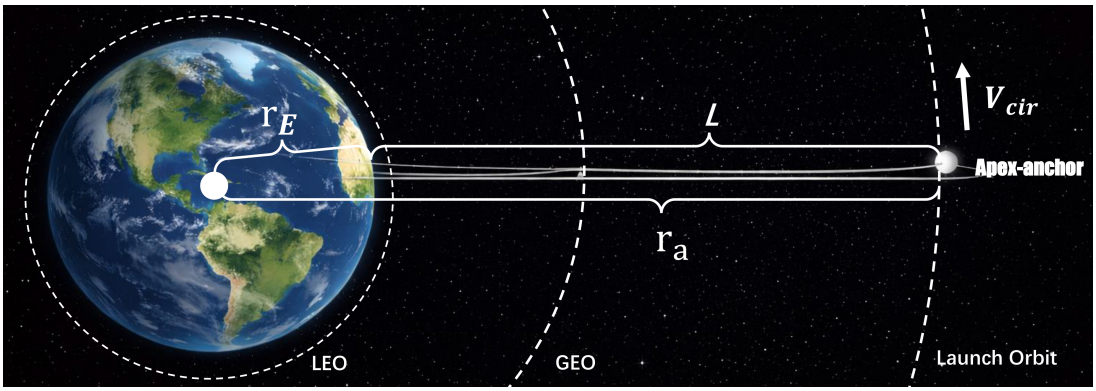


Figure 4: The diagram of Apex-anchor orbit and V_{cir}

After reaching the apex anchor, all materials are transferred to the Moon. Let the tether length be $L = 100{,}000\,\text{km}$, r_E represent as the radius of the Earth, then define the radius of the apex anchor in Figure 4 as

$$r_a = r_E + L$$

To maintain analytical tractability, the apex anchor is approximated as an equivalent circular parking point at radius r_a. The corresponding circular velocity is

$$v_{\text{cir}}(r_a) = \sqrt{\frac{\mu}{r_a}}$$

where μ is the Earth's standard gravitational parameter. A Hohmann transfer from r_a to the Moon's orbital radius r_M is then constructed, with semi-major axis $a = (r_a + r_M)/2$. The required velocity respect to r_a is

$$v_{\text{trans}}(r_a) = \sqrt{\mu\left(\frac{2}{r_a} - \frac{1}{a}\right)}$$

Accordingly, the rocket velocity increment required for the apex-to-Moon pace is approximated by

$$\Delta v_a = v_{\text{trans}}(r_a) - v_{\text{cir}}(r_a)$$

Using the Tsiolkovsky rocket equation, the velocity increment required at launch site i satisfies

$$\Delta V_i = V_e \ln \frac{m_0}{m_1} \tag{4}$$

where v_e denotes the effective exhaust velocity, m_1 represents the dry mass regarded as 125 ton, and m_0 is the total mass including propellant. Then, we obtain $m_0 = m_1 e^{\Delta v_a / v_e}$ and the corresponding propellant m_p

$$m_p = m_0 - m_1 = m_1 \left(e^{\Delta V_i / V_e} - 1\right) \tag{5}$$

Then the cost per rocket launch from the apex anchor can also be represented as:

$$C_{\text{launch},a} = k(m_0 - m_1) = k\, m_1 \left(e^{\Delta v_a / v_e} - 1\right)$$

Let N_a denote the total number of apex-to-Moon launches required to deliver D_{total} tons of material. The total rocket cost for the apex-to-Moon part is then

$$C_{r,a} = N_a(C_{\text{launch},a} + C_0) \tag{6}$$

Combining two components, the total cost under the elevator-only scenario is

$$C_{\text{e,total}} = C_{e,\text{lift}} + C_{r,a}$$

It serves as an environmentally clean yet capacity-limited benchmark, highlighting the trade-off between reduced fuel consumption and constrained throughput.

5.1.2 Rocket-Only Delivery Model

In the rocket-only scenario, all the construction materials and resources are transported from the Earth port to the Moon directly using traditional rocket launches. Let D_{total} denote the total mass required for constructing the Moon Colony. In this scenario,

$$m_r = D_{\text{total}} = 10^8 \text{tons}, \qquad m_e = 0$$

All the rockets must first reach a Low Earth Orbit (LEO). Earth rotation provides an initial inertial velocity component at the i-th launch site:

$$V_{\text{LEO}} = 7.8 \text{ km/s}, \qquad V_{p,i} = V_{\text{eq}} \cos \phi_i, \qquad V_{\text{eq}} = 466 \text{ m/s}$$

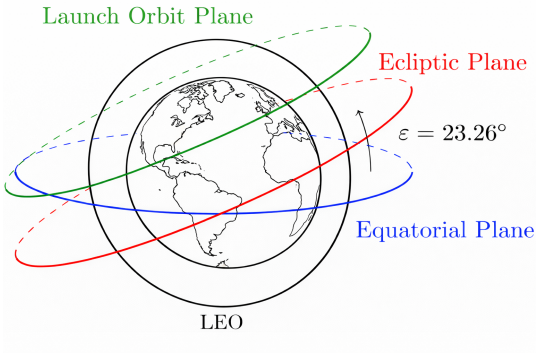
We lump gravity and aerodynamic losses into a constant Δv_{loss}. Hence, the required velocity increment from the ground to LEO is approximated by

$$\Delta \vec{V}_{g \to \text{LEO},i} = \vec{V}_{\text{LEO}} - \vec{V}_{p,i} + \Delta \vec{V}_{\text{loss}}, \tag{7}$$

After reaching LEO, the rocket performs translunar injection (TLI) to enter the Earth–Moon transfer orbit. We adopt the TLI speed increment. Launch sites are classified by latitude relative to 23.26° in Figure 5, which determines whether an inclination penalty is required.



(a) two launch sites categories



(b) Diagram of the geometric relationship between the Equatorial Plane, Ecliptic Plane, and Launch Orbit Plane (not to scale)

Figure 5: Global launch sites and the geometric mechanism

For the i-th launch site located at latitude ϕ_i, an inclination penalty is defined as

$$\theta_i = \begin{cases} \phi_i - 23.26°, & \phi_i > 23.26° \\ 0, & \phi_i \leq 23.26° \end{cases}$$

Using such a penalty, the required velocity increment can be approximated by

$$\Delta V_i = \sqrt{V_{\text{LEO}}^2 + V_{\text{Trans}}^2 - 2V_{\text{LEO}}V_{\text{Trans}}\cos\theta_i} \tag{8}$$

where $V_{\text{Trans}} = 3067$ m/s is the characteristic translunar injection velocity.

Based on the Tsiolkovsky rocket equation (Equation 4), the propellant mass m_p (Equation 5) is generated. Assuming the rocket propellant cost is proportional to the consumed propellant mass, the cost per launch at site i is modeled as:

$$C_{\text{launch},i} = k\,(m_0 - m_1) = k\,m_1\left(e^{\Delta V_i/V_e} - 1 + C_0\right)$$

where k denotes the price of rocket propellant and C_0 represents the depreciation costs.

To facilitate comparison and scheduling among multiple launch sites, an efficiency coefficient is further introduced as follows, and all launch sites are ranked in descending order of η_i.

$$\eta_i = \frac{1}{1 + \theta_i/100}$$

Let F_i and P_i denote the annual launch frequency and payload capacity of launch site i, respectively. The decision variable N_i represents the total number of launches assigned to site i, representing as:

$$\forall\, 0 \leq N_i \leq F_i T_r, \qquad \sum_{i=1}^{10} N_i P_i = m_r$$

A greedy allocation strategy is adopted, whereby launch sites with higher efficiency is assigned to launches up to their capacity limits before lower-efficiency sites are utilized. Since launches across different sites can be conducted in parallel, The total transportation time is determined by the most heavily utilized site:

$$T_r = \max\left(\frac{N_i}{F_i}\right) \tag{9}$$

The total rocket transportation cost is therefore given by

$$C_{\mathrm{r}} = \sum_{i=1}^{10} N_i C_{\text{launch},i} \tag{10}$$

5.2 Hybrid Optimization Delivery Model

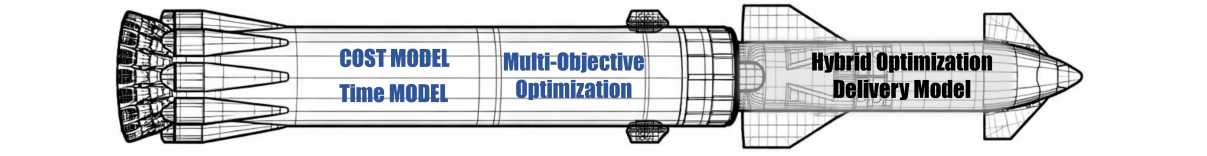


Figure 6: Rocket diagram for optimization model

Therefore, a hybrid delivery model combining rockets and the space elevator is introduced. Let $\lambda \in [0, 1]$ represent the proportion of total mass transported through the space elevator. Therefore, the mass allocations of the two transportation methods are defined as

$$m_e = \lambda D_{\text{total}}, \qquad m_r = (1 - \lambda) D_{\text{total}}$$

Time Model

For the hybrid strategy, the elevator-side delivery time follows the capacity constraint in the elevator-only scenario (Equation 2), while the rocket-side delivery time follows the scheduling rule of the rocket-only model (Equation 9). The overall delivery time is therefore given by:

$$T_e(\lambda) = \frac{m_e}{c_E} = \frac{\lambda D_{\text{total}}}{c_E}, \qquad T_r(\lambda) = \max\left(\frac{N_i(\lambda)}{F_i}\right) \tag{11}$$

As the space elevator and Earth-based rocket systems operate simultaneously, the delivery time under the hybrid is determined by the slower of the two processes:

$$T(\lambda) = \max\left(T_e(\lambda),\ T_r(\lambda)\right)$$

Cost Model

Under the hybrid scenario, the total cost consists of three components: the Earth-based rocket launch cost, the elevator lifting cost, and the apex-to-Moon rocket cost associated with elevator deliveries.

For the hybrid strategy, the total cost consists of three components: Earth-based rocket launch costs summed over all sites (Equation 10), elevator lifting costs proportional to the delivered mass (Equation 3), and apex-to-Moon transfer costs for elevator-delivered payloads (Equation 6), where $N_a(\lambda)$ denotes the number of apex-to-Moon launches required to deliver m_e tons. The three components are given by:

$$C_r(\lambda) = \sum_{i=1}^{10} N_i(\lambda)\, C_{\text{launch},i}, \quad C_{e,\text{lift}}(\lambda) = C_e \lambda D_{\text{total}}, \quad C_{r,a}(\lambda) = N_a(\lambda) C_{\text{launch},a}$$

Combining all cost components, the total transportation cost under the hybrid delivery strategy is expressed as

$$C(\lambda) = C_{e,\text{lift}}(\lambda) + C_{r,a}(\lambda) + C_r(\lambda) \tag{12}$$

Multi-Objective Optimization

To balance cost and time efficiency, both parameters are normalized using the two baseline scenarios as reference bounds. The normalized cost and time are defined as

$$C_{\text{norm}}(\lambda) = \frac{C(\lambda) - min\, C(\lambda)}{max\, C(\lambda) - min\, C(\lambda)}, \qquad T_{\text{norm}}(\lambda) = \frac{T(\lambda) - min\, T(\lambda)}{max\, T(\lambda) - min\, T(\lambda)}$$

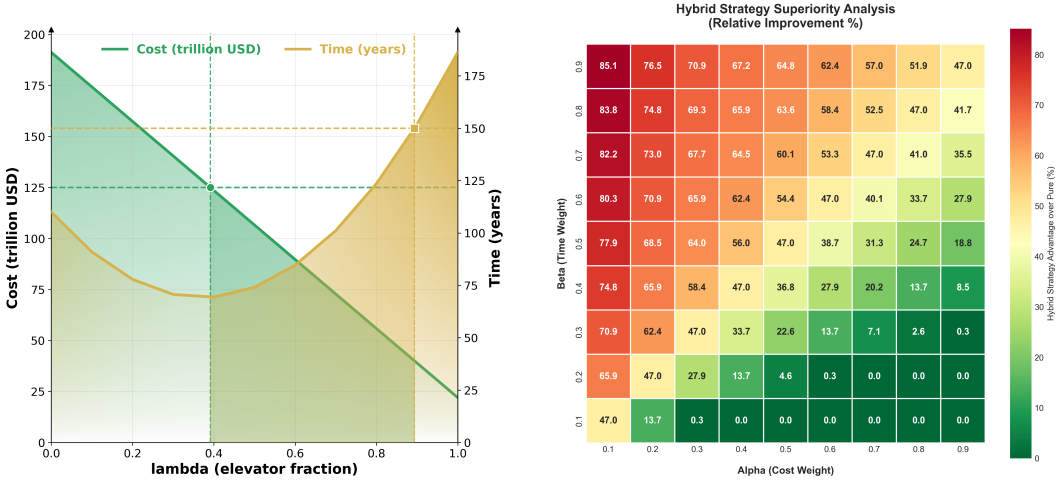
Where α and β represent the relative importance of cost and time, respectively, a composite objective function is then constructed.

$$J(\lambda) = \alpha\, C_{\text{norm}}(\lambda) + \beta\, T_{\text{norm}}(\lambda), \qquad \alpha + \beta = 1$$

By minimizing the $J(\lambda)$ over the feasible interval $\lambda \in [0, 1]$, the optimal hybrid delivery ratio λ' can be identified, providing a balanced delivery strategy for constructing the Moon Colony.

5.3 Results and Key Findings

The numerical results of Task I reveal clear trade-offs between delivery time and total cost across the three delivery strategies. The results show that varying the allocation ratio λ reveals clear trade-offs between cost and completion time.



(a) Cost–time trade-off as a function of λ under cost or time constraints.

(b) Relative performance improvement of the hybrid strategy under different preferences

Figure 7: Cost–time trade-off and robustness of the hybrid allocation strategy

Figure 7(a) shows the λ-driven trade-off: higher ratios reduce costs but increase delivery time via elevator bottlenecks. Consequently, cost/time constraints restrict the feasible λ interval. Figure 7(b) maps the hybrid strategy's performance across the full

Table 3: Performance comparison of three delivery schemes

Scheme	λ^*	Time (years)	Total Cost (USD)	$J(\lambda)$
Pure Space Elevator	1.000	186.2197	2.2000×10^{13}	0.4500
Pure Rocket	0.000	109.5890	1.9126×10^{14}	0.7073
Best Hybrid (λ)	0.617	86.9778	8.6823×10^{13}	0.2796

(α, β) preference space. The hybrid approach consistently outperforms pure strategies, yielding >40% improvement in balanced regimes and >80% under time-priority settings, confirming its robustness.

Table 3 summarizes the representative outcomes, when $\alpha : \beta$ fixed at 0.55 : 0.45. The pure space elevator strategy ($\lambda = 1$) minimizes total cost at $\$2.20\times10^{13}$ but requires over 180 years to complete, while the pure rocket strategy ($\lambda = 0$) shortens the delivery time to about 110 years at a much higher cost of $\$1.91\times10^{14}$.

The hybrid strategy provides a balanced alternative. At the optimal allocation $\lambda^* = 0.617$, the completion time is reduced to approximately 87 years and the total cost to $\$8.68\times10^{13}$. The corresponding normalized objective value $J(\lambda^*) = 0.2796$ represents a 37.8% improvement over the elevator-only case and a 60.5% improvement over the rocket-only case, confirming the hybrid architecture as the preferred solution.

6 Task II: Non-Ideal Conditions & Robustness Analysis

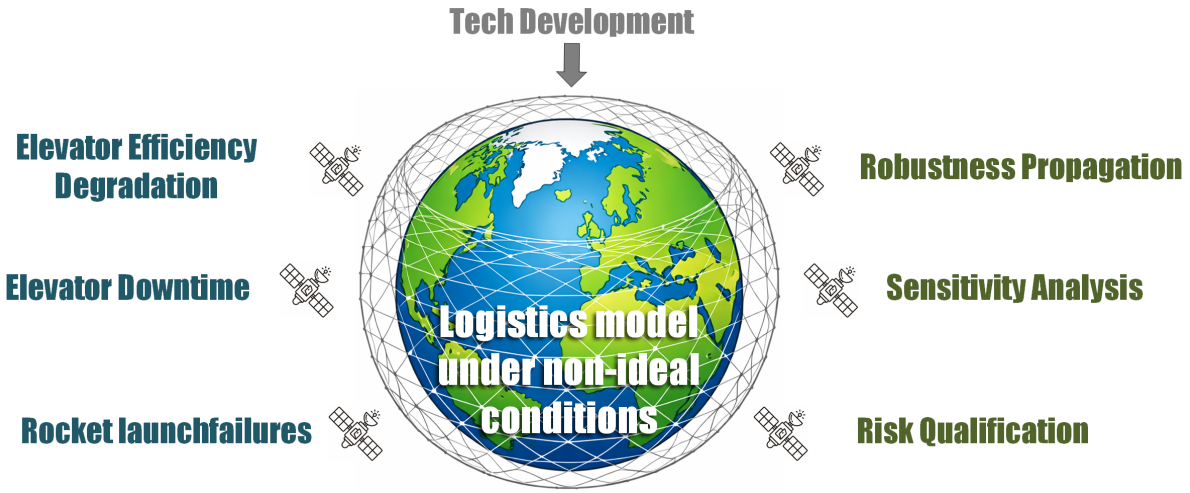


Figure 8: The Earth diagram of the non-ideal conditions and Robustness Analysis

The section extends the fixed model into a robustness analysis framework. By introducing disruptions and time-varying reliability, decision-makers can qualify how the resulting uncertainty propagates into the total time $T(\lambda)$ and cost $C(\lambda)$.

6.1 Sources of System Perturbations

We model three dominant non-ideal conditions' mechanisms that directly affect through [5, 2]:

- Elevator efficiency degradation, including tether sway and operational throttling
- Temporary elevator downtime

- Rocket launch failure

This study introduces the elevator efficiency factor $\xi_{e,t}$, the elevator availability indicator $\delta_{e,t}$, and the rocket single-launch failure probability $p_{f,t}$ as follow. All three parameters are normalized to lie in the interval (0,1) because they represent probabilistic or fractional system states rather than absolute quantities.

$$\xi_{e,t} \in (0,1), \qquad \delta_{e,t} \in \{0,1\}, \qquad p_{f,t} \in (0,1)$$

• Source A: Space elevator efficiency degradation

Even when the elevator is operational, safety constraints can reduce its effective rate. $\xi_{e,t}$ denote the multiplicative efficiency factor in period t. To capture nominally and occasionally degraded behavior while securing the physical bound $(0,1)$, The factor is described by the Beta-type efficiency model, which satisfies the bound check and allows asymmetric allocations concentrated around 1.

$$\xi_{e,t} \in (0,1), \qquad \xi_{e,t} \sim \beta(\alpha_e, \beta_e) \tag{13}$$

• Source B: Temporary elevator downtime

Space elevator downtime is modeled as temporary unavailability rather than a permanent loss. We define an availability indicator as follows. When $\delta_{e,t} = 0$, the elevator contributes no capacity during period t.

$$\delta_{e,t} \in \{0,1\}, \qquad \Pr(\delta_{e,t}) = \begin{cases} p_{\text{up}}(t), & \delta_{e,t} = 1 \\ 1 - p_{\text{up}}(t), & \delta_{e,t} = 0 \end{cases} \tag{14}$$

• Source C: Rocket launch failures

Rocket launch failures are modeled as probabilistic events that result in no payload delivery, increasing both delivery time and cost. Let $p_{f,t} \in (0,1)$ denote the single-launch failure probability in period t. Assuming independent outcomes across launches, the expected fraction of successful launches is given by $(1 - p_{f,t})$, which can serve as a multiplicative discount coefficient.

$$p_{f,t} \in (0,1) \tag{15}$$

Time Evolving Reliability Adjustment

In practical application, a constant failure rate is not realistic. Reliability may improve due to multiple reasons. Therefore, three exponential decay forms are used to describe the failure probabilities:

$$p_{f,t} = p_{f,0}\, e^{-k_f t}, \qquad p_{\text{down}}(t) = p_{d,0}\, e^{-k_e t}, \qquad p_{\text{up}}(t) = 1 - p_{\text{down}}(t) \tag{16}$$

where $p_{f,0}$ and $p_{d,0}$ are initial failure and unavailability degree, while k_f and k_e represent the Respective growth rates of rockets and elevator operations.

6.2 Robustness Propagation to Time and Cost

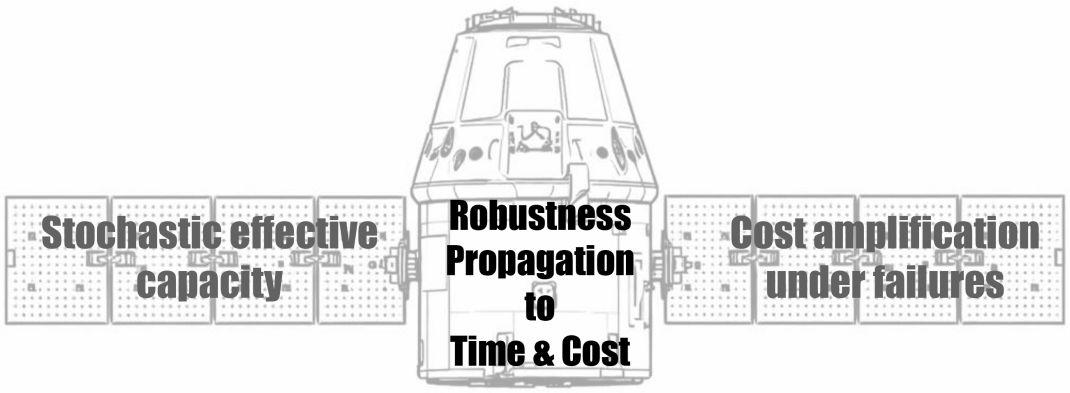


Figure 9: Satellite diagram for robust analysis of time and cost

Stochastic effective capacity

From Task I, the total elevator lifting capacity is c_E (Equation 3). Under non-ideal operations, the realized elevator capacity in period t is systematically discounted by both availability and efficiency:

$$c_{E,t}^{\text{eff}} = \delta_{e,t}\, \xi_{e,t}\, c_E \tag{17}$$

Hence, the elevator-side completion time becomes a random variable:

$$T_e(\lambda) = \frac{m_e}{c_E^{\text{eff}}} \quad \text{(annualized form)} \tag{18}$$

where c_E^{eff} denotes the total effective capacity over the whole planning period.

For the rocket subsystem, Task I determines the nominal completion time $T_r(\lambda)$ (Equation 11). When launch fails, the effective delivered payload per launch is statistically reduced in expectation by $(1 - p_{f,t})$. Thus, to deliver the same target mass, the required launches with approximately a factor $1/(1 - p_{f,t})$, which substantially increases the schedule pressure on the bottleneck site:

$$T_r^{\text{eff}}(\lambda) \approx \max\left(\frac{N_i(\lambda)}{F_i(1 - p_f)}\right) \tag{19}$$

Cost amplification under failures

Task I expresses total hybrid delivery cost C as a combination of two transportation components (Equation 12). When the rocket fails, additional launches are needed to compensate for its mistake, increasing both $C_r(\lambda)$ and the apex-to-Moon rocket cost $C_{r,a}(\lambda)$. A first-order expectation correction is:

$$C_r^{\text{eff}}(\lambda) \approx \frac{1}{1 - p_f} \sum_{i=1}^{10} N_i(\lambda)\, C_{\text{launch},i}, \qquad C_{r,a}^{\text{eff}}(\lambda) \approx \frac{1}{1 - p_f}\, C_{r,a}(\lambda) \tag{20}$$

Elevator downtime primarily impacts time. Actually, it may also introduce maintenance costs in practice, but due to Task I models elevator lifting cost as linear in m_e, we keep $C_{e,\text{lift}}(\lambda)$ unchanged and treat downtime.

Figure 10 shows that combined failures (All Risks) induce the largest outward shift, indicating simultaneous cost inflation and schedule delay.

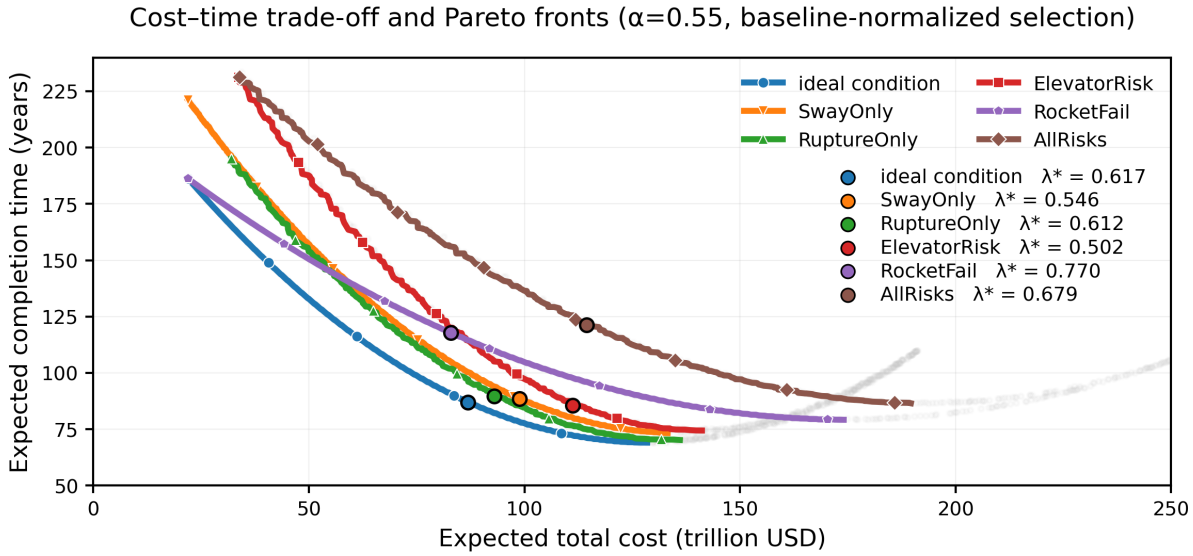


Figure 10: Cost-time trade-off curves under six non-ideal scenarios

6.3 Sensitivity Analysis and Risk Metrics

In a bid to quantify the outcomes of non-ideal issues, we introduce robustness based on two dimensions, parameter sensitivity and tail-risk metrics.

Parameter sensitivity

We disturb the most vital parameters and observe the outcomes in $T(\lambda)$ and $C(\lambda)$:

Table 4: Parameters perturbed in sensitivity analysis.

Component	Parameter(s)	Meaning
Elevator efficiency	(α_e, β_e)	Beta distribution of $\xi_{e,t}$
Elevator availability	$(p_{d,0}, k_e)$	Downtime probability decay
Rocket reliability	$(p_{f,0}, k_f)$	Launch failure decay

Each parameter is varied within a stress range while holding others fixed, and the outcomes are recorded, as follow:

$$\frac{\Delta T(\lambda)}{T(\lambda)} \qquad \frac{\Delta C(\lambda)}{C(\lambda)}$$

It represents the dominant contributor and identifies the reliability threshold value when the system becomes unstable. Beyond parameter perturbations, we also test decision robustness against preference shift by varying the weight α in the composite objective.

Figure 11 summarizes how the optimal policy λ^* changes across scenarios; frequent switching implies that the recommended allocation is preference-sensitive rather than structurally stable.

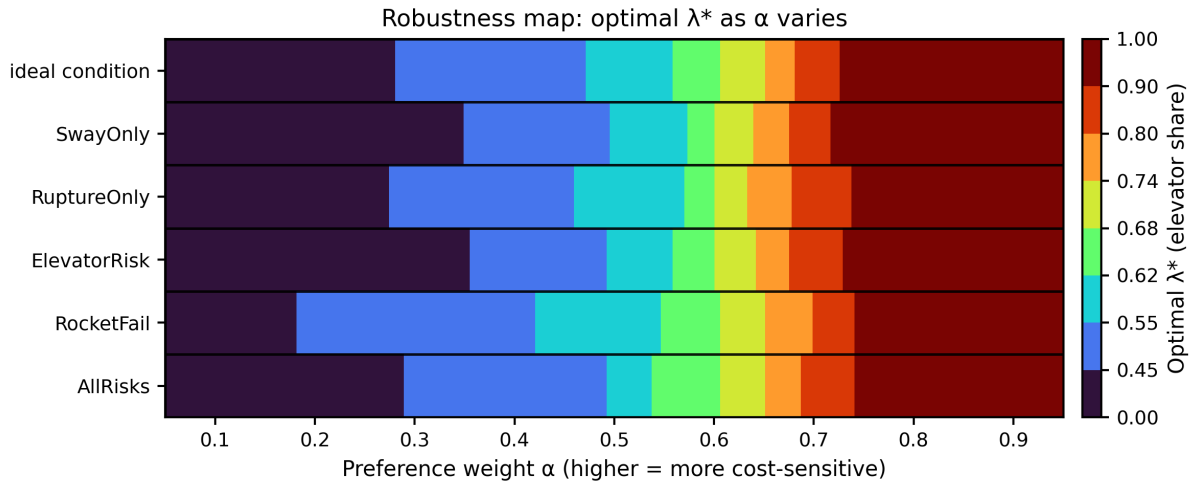


Figure 11: Robustness map of the optimal elevator share λ^* under preference shift

Tail-risk metrics

Rare failures and tiny errors may accumulate over decades, which require further assessment rather than mean performance. Thus, we adopt three robustness metrics:

$$\mathbb{E}[T(\lambda)] = \frac{1}{S}\sum_{s=1}^{S} T^{(s)}(\lambda)$$

$$\mathrm{CVaR}_\alpha(T(\lambda)) = \mathbb{E}[T(\lambda) \mid T(\lambda) \geq \mathrm{VaR}_\alpha]$$

$$P_{\text{collapse}}(\lambda) = \Pr\big(T(\lambda) > T_{\text{fail}}\big)$$

where S is the number of Monte Carlo scenarios, α is a tail level, and T_{fail} is the deadline of mission-defined unacceptable completion. The first metric captures average schedule, the second captures extreme-delay risk, and the third quantifies the probability that the plan fails to meet a hard schedule constraint.

Table 5: Performance degradation with the policy fixed at the baseline optimum ($\lambda = 0.617$)

Scenario	**Time (years)**	**ΔTime (years)**	**ΔCost (trillion USD)**
Ideal condition	87.0	0.0	0.00
Sway-only	100.4	+13.4	0.00
Rupture-only	90.7	+3.8	+6.09
Elevator Risk	104.8	+17.9	+7.44
Rocket Fail	91.2	+4.2	+36.60
All Risks	109.1	+22.1	+44.06

Table 5 evaluates the robustness of a fixed baseline policy. Elevator-side disruptions mainly affect completion time, while rocket failures dominate cost inflation. Under combined failures, both time and cost deteriorate markedly, indicating the limited robustness of a fixed allocation.

6.4 Monte Carlo Simulation Procedure

For a fixed allocation λ, system reliability is sampled period-by-period, and Monte Carlo simulations are used to estimate the distributions of completion time and cost.

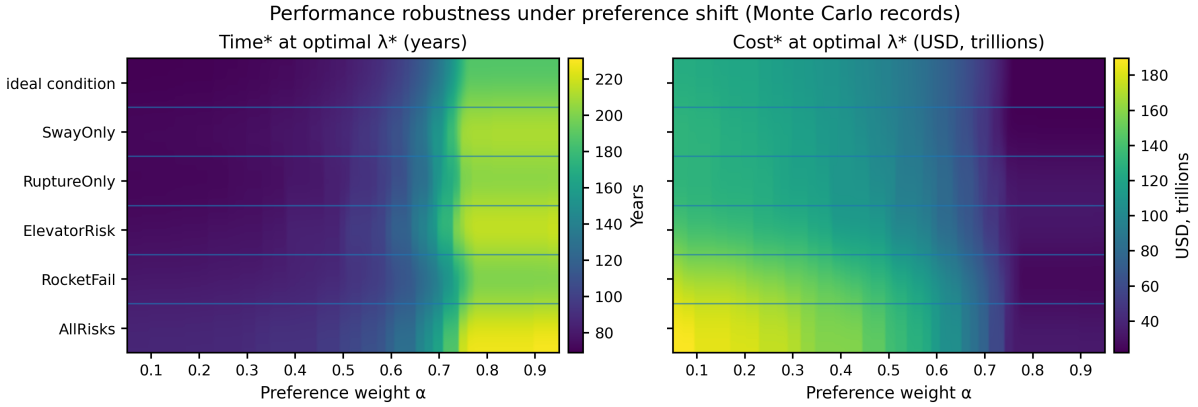


Figure 12: Performance robustness under preference shift

Figure 12 shows the expected time and cost at the optimal policy $\lambda^*(\alpha)$, where darker colors indicate worse performance. High-intensity regions under compounded risks indicate limited robustness.

To further quantify the sensitivity of the optimal allocation to preference shifts, We define the Preference Stability Index (PSI) as

$$\text{PSI} = \frac{1}{N-1} \sum_{i=1}^{N-1} |\lambda^*(\alpha_{i+1}) - \lambda^*(\alpha_i)| \tag{21}$$

where $\{\alpha_i\}_{i=1}^{N}$ denote the sampled preference weights. A smaller PSI indicates greater stability of the optimal allocation with respect to preference shifts. In our simulations, the computed PSI is only 2.5%, indicating that the optimal allocation varies smoothly across a wide range of preference weights and remains structurally stable. This low sensitivity demonstrates that the proposed hybrid strategy maintains robustness against preference uncertainty, providing a reliable decision rule under non-ideal conditions.

7 Task III: Water Supply Sustainability Model

Through the previous analysis, the cislunar delivery system has been obtained, which measures the cost of transporting resources. Additionally, as a necessity of sustainable development, water resource is regarded as the most significant good in moon colony [7]. Therefore, this section develops a decision-oriented water supply model as a case study. Linking directly to the previous cislunar logistics framework, it can quantify the water demand of the Moon Colony.

7.1 Problem Formulation and System Dynamics

Water supply is modeled as a balance system over regular decision periods. Let W_t represent the available water storage at the beginning of period t. During each period, water is consumed by the residents on the Moon Colony, partially recovered through recycling, and supplied through external delivery.

The total water consumption in period t is proportional to population size,

$$W_t^{\text{con}} = \omega \cdot P \tag{22}$$

where ω is the average individual water consumption rate and $P = 100{,}000$ is the colony population.

A fraction of consumed water is recovered through a closed-loop recycling system. Let $\eta_w \in (0, 1)$ represent the recycling efficiency. The recovered water in period t is

$$W_t^{\text{rec}} = \eta_w W_t^{\text{con}} \tag{23}$$

Let W_t^{in} represent externally delivered water in period t. Therefore, the water storage dynamics are shown as:

$$W_{t+1} = W_t - W_t^{\text{con}} + W_t^{\text{rec}} + W_t^{\text{in}} \tag{24}$$

In a bid to ensure system feasibility, the following failure constraint is defined:

$$\forall t \in N,\ W_t \geq W_{\min} \tag{25}$$

where $W_{\min}$ is the minimum safe water reserve. If beyond the boundary of constraints, the colony is regarded as non-operational and the delivery strategy is deemed infeasible. This description captures the physical processes that secure water availability while remaining practical with the previously defined logistics capacity constraints.

7.2 Decision-Based Water Allocation Framework (MDP-based)

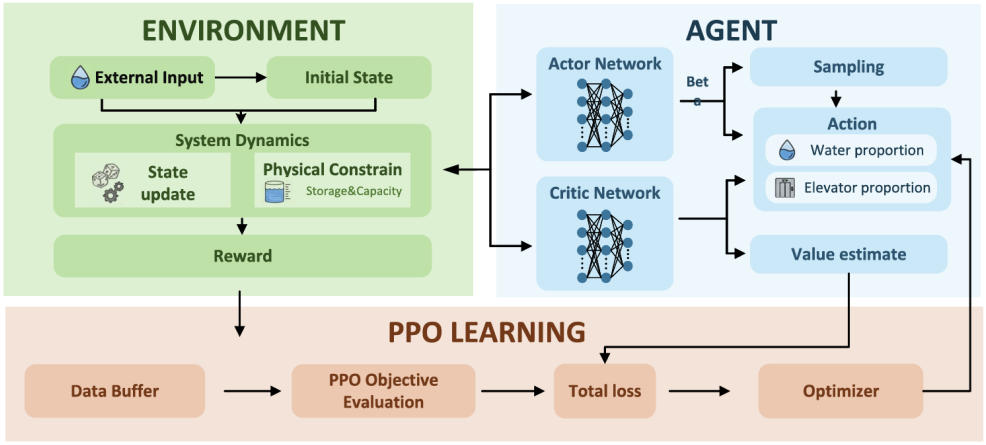


Figure 13: Illustration for the PPO process

The water supply problem involves sequential decisions under uncertainty, including variability in delivery capacity and system reliability. It can be formulated at a high level as a Markov decision process (MDP) [7, 8].

The system state combines decision-relevant information of the overall delivery system and Moon Colony, as follow:

- remaining water demand
- current water storage level
- the operational condition

Control actions determine how much of the available transportation capacity is allocated to water delivery in each period. Rather than optimizing short-term delivery efficiency alone, decisions must balance three competing objectives for a long period:

maintaining water safety, sustaining construction progress, and avoiding system failure.

Additionally, the decision-making process evaluates actions based on their impact on water feasibility and delivery performance. The reward design intuitively reflects these priorities:

- policies are encouraged to maintain adequate water reserves
- penalized for water shortages or system collapse
- steady progress in material delivery is favored

This structure ensures conservative water management under uncertain operating conditions without requiring explicit optimization for short periods.

7.3 Algorithmic Solution and Policy Output

To solve the resulting constrained sequential decision problem with continuous control variables, a reinforcement learning (RL) approach based on Proximal Policy Optimization (PPO) is adopted. PPO is well-suited for stochastic environments and continuous decision spaces. It provides stable policy learning without requiring linearization for an explicit system, as shown in Figure 13.

The learned policy produces an estimate of the external water supply, which is required to maintain long-term feasibility. Specifically, the algorithm outputs the annual water shortfall

$$W_{\text{gap}} = \sum_{t} \max\left(0,\, W_{\min} - W_t\right) \tag{26}$$

which represents the minimum external water input needed to avoid system failure.

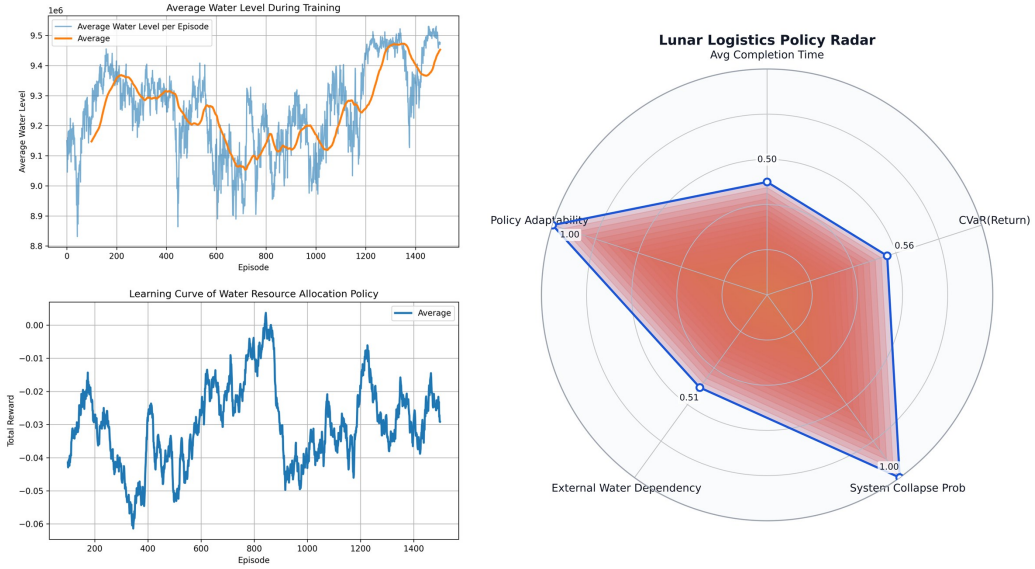


Figure 14: Results of the task three model

This annual water demand is converted into an equivalent transportation mass and incorporated into the hybrid delivery model established in Task I and II. It acts as a case study of an additional logistics requirement. As a result, water sustainability directly affects total delivery cost, completion time, and the optimal allocation between two scenarios of rockets and the space elevator. It ensures full integration of life-support constraints into the overall system-level decision framework.

Figure 14 demonstrates the performance of the model, which is skilled in maintaining the Moon Colony, promising the water usage on the moon. Meanwhile, the other three aspects have gained a median score of about 50%. The other two diagrams for the training process showcase the changes in water resource reserves, which is another proof of the strong ability to keep the water supply.

8 Task IV: Environmental Impact Assessment Model

This section evaluates the environmental impact of the proposed delivery system and integrates environmental factors into the existing decision framework, which supports sustainable planning [9].

We compare three delivery scenarios: Earth-based rocket only, space elevator only, and the hybrid system parameterized by the cargo allocation ratio λ. In that case, environmental impacts can be assessed on a consistent basis and directly linked to strategic choices.

8.1 Environmental Impact Modeling

This subsection extracts environment impact factors from underlying delivery mechanisms and combines those sets of small, ready components into a unified impact framework.

• Rocket-Induced Environmental Impact Component

For Earth-based launch, the primary environmental burden mainly comes from repeated launches. More frequent launches undoubtedly increase the environmental stress. Let $N_r(\lambda)$ represent the total number of rocket launches with respect to the previously given allocation ratio λ. Let e_r denote the effective emission intensity per launch. The rocket-induced environmental impact is modeled as

$$E_r(\lambda) = e_r \, N_r(\lambda) \tag{27}$$

To account for non-ideal conditions and launch failures from Task II, we introduce a reliability multiplier $\kappa_r \geq 1$, adjusting the rocket.

$$\widetilde{E}_r(\lambda) = \kappa_r \, E_r(\lambda) \tag{28}$$

• Space-Elevator Energy Consumption Component

For the space elevator system, environmental impact is dominated by electricity consumption rather than direct atmospheric emissions. For the launches from the apex anchor, the rockets are not in the atmosphere, whose environmental impact can be neglected. Let $m_e(\lambda)$ denote the total mass transported by elevators. Let ε_e be the energy intensity per unit mass and g the electricity carbon intensity. The elevator-induced environmental impact is modeled as

$$E_e(\lambda) = g \, \varepsilon_e \, m_e(\lambda) \tag{29}$$

Similarly, elevator downtime and operational inefficiencies are captured through a multiplier $\kappa_e \geq 1$, giving

$$\widetilde{E}_e(\lambda) = \kappa_e \, E_e(\lambda) \tag{30}$$

Unified Environmental Impact Function

Combining two contributions, we define a unified environmental impact function

$$E(\lambda) = \widetilde{E}_r(\lambda) + \widetilde{E}_e(\lambda) \tag{31}$$

Allocation $\lambda \in [0, 1]$ represents not only the hybrid strategy, but also Rocket-only and elevator-only strategies at boundary cases when $\lambda = 0$ and $\lambda = 1$, respectively.

8.2 Environment-Augmented Trade-off Model

Building upon the previous time-cost trade-off model established in Task I - II, this step augments environmental impact as an additional optimization dimension [3]. Let $T(\lambda)$ and $C(\lambda)$ denote the completion time and total monetary cost, with normalized forms $T_{\text{norm}}(\lambda)$ and $C_{\text{norm}}(\lambda)$ defined previously. We define the normalized environmental metric as

$$E_{\text{norm}}(\lambda) = \frac{E(\lambda) - \min E(\lambda)}{\max E(\lambda) - \min E(\lambda)} \tag{32}$$

The environment-augmented objective function is then expressed as

$$J_{\text{aug}}(\lambda) = \alpha C_{\text{norm}}(\lambda) + (1 - \alpha) T_{\text{norm}}(\lambda) + \gamma E_{\text{norm}}(\lambda), \qquad \lambda \in [0, 1] \tag{33}$$

where α retains its original interpretation as the cost–time preference parameter, and $\gamma \geq 0$ represents the relative importance assigned to environmental impact.

For robustness analysis, the time component $T(\lambda)$ may be replaced by the risk-sensitive metrics $\mathbb{E}[T(\lambda)]$, $\text{CVaR}_\alpha(T(\lambda))$, and $P_{\text{collapse}}(\lambda)$ defined in Task II, which may not alter the structure of the augmented objective Increasing environmental priority systematically shifts the optimal allocation toward higher elevator utilization.

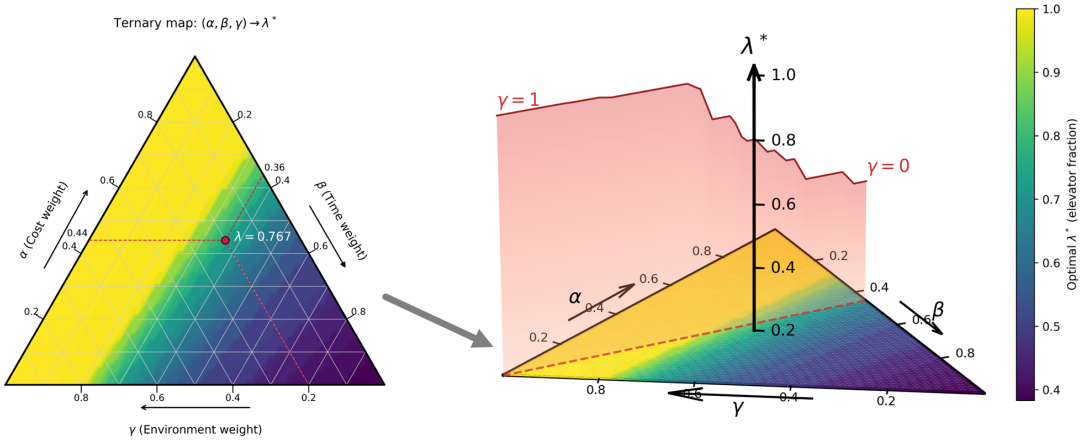


Figure 15: Preference-to-decision sensitivity of the environment-augmented objective

Figure 15 illustrates how preference weights determine the optimal cargo allocation under the environment-augmented objective. In the left panel, the ternary map shows the mapping $(\alpha, \beta, \gamma) \to \lambda^\star$, with color indicating the optimal elevator fraction.

The highlighted red point serves as an example of a specific preference combination and its corresponding optimal allocation. The right panel fixes the cost–time ratio at $\alpha : \beta = 0.55 : 0.45$ and varies the environmental weight γ, showing that increasing environmental priority systematically shifts the optimal solution toward higher elevator utilization while preserving the previously identified time–cost balance.

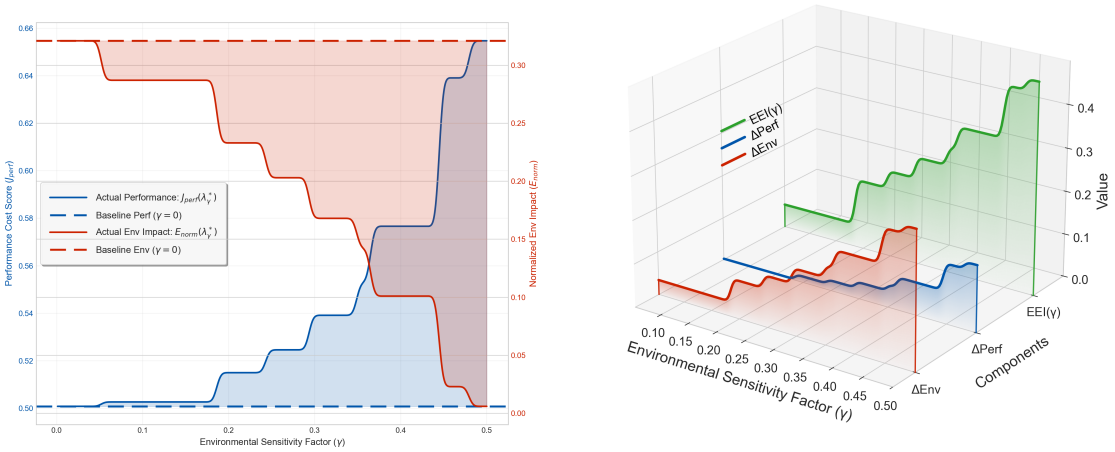
8.3 Scenario-Based Strategy Analysis

To evaluate the efficiency of incorporating environmental considerations, we introduce an *Environmental Efficiency Index* (EEI), which measures the additional system-level cost required to achieve a unit reduction in normalized environmental impact. The EEI is defined as

$$\mathrm{EEI}(\gamma) = \frac{J_{\mathrm{aug}}(\lambda^*_\gamma) - J(\lambda^*_{\gamma=0})}{E_{\mathrm{norm}}(\lambda^*_{\gamma=0}) - E_{\mathrm{norm}}(\lambda^*_\gamma)} \tag{34}$$

where $\lambda^*_{\gamma=0}$ denotes the optimal allocation obtained under the original time–cost objective, and λ^*_γ is the optimal allocation under environmental weight γ. A smaller EEI indicates that environmental improvement is achieved at a lower performance cost.

Figure 16 shows the trade-off between the objective value J and environmental impact E_{norm} as functions of λ (left), and the corresponding $\mathrm{EEI}(\gamma)$ curve summarizing the efficiency of environmental improvement across different weights (right).



(a) Cost–time trade-off under varying γ

(b) Component responses to γ

Figure 16: Environmental sensitivity analysis of the hybrid delivery strategy

Rather than relying on a single preference setting, we evaluate EEI over a decision-relevant range of environmental weights $\gamma \in [\gamma_{\min}, \gamma_{\max}]$, where environmental considerations meaningfully influence the optimal allocation. To summarize performance over this range, we compute the interval-average EEI

$$\overline{\mathrm{EEI}} = \frac{1}{\gamma_{\max} - \gamma_{\min}} \int_{\gamma_{\min}}^{\gamma_{\max}} \mathrm{EEI}(\gamma)\, \mathrm{d}\gamma \tag{35}$$

which captures the average cost efficiency of environmental improvement.

The EEI over $\gamma \in [0.05, 0.50]$ is 0.222, indicating that the hybrid strategy achieves environmental improvement at a low system-level cost. Relative to the rocket-only or elevator-only configuration, the hybrid approach delivers a substantial reduction in environmental impact without a disproportionate increase in time or cost. As a result, the EEI results confirm the hybrid strategy as the most efficient and balanced option under the integrated time–cost–environment framework.

9 Model Evaluation and Further Discussion

9.1 Strengths

Our model offers the following strengths:

- The proposed framework integrates cost, time, risk, and environmental impact within a unified optimization structure.
- A comprehensive evaluation model for logistics analysis is scientifically sound. We combined the method of mechanism analysis and comprehensive evaluation method to obtain a more objective evaluation model.
- Our model effectively achieves all the objectives, can handle large amounts of data and has the flexibility and robustness we expect.

9.2 Weaknesses &Further Discussion

Our model has the following limitations and related improvements:

- There are limits to the parameters we assigned, as the real data can not be collected now. Hence, the results are all based on the expected situation.
- In our hypothesis, we assumed that the decision concentration of government is neutral. However, the real condition is complex, leading to a comprehensive set to the data.
- While the model captures time evolution, strategic decisions are represented in an aggregated manner. Future work may introduce multiple stakeholders and preference heterogeneity to better reflect real-world policy interactions.

10 Conclusion

This paper presents a quantitative framework for evaluating hybrid delivery strategies for constructing Moon Colony [10]. By comparing Earth-based rocket-only, Space Elevator System-only, and hybrid delivery scenario, the model represents that a hybrid strategy achieves the balance among total cost, construction time, and operational robustness. Sensitivity analysis indicates that this strategy remains stable under reasonable parameter variations in non-ideal conditions. In addition, the water sustainability analysis highlights the importance of closed-loop water recycling supplemented by periodic external supply. Overall, the proposed environment-augmented framework provides a concise, transparent, and decision-oriented basis for planning large-scale lunar infrastructure.

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Plane for Moon Colony!

To: All staffs
From: Team # 2600308
Date: 2026/2/3
Subject: A Feasible Scheme For Moon Colony Base

Dear MCM Agency,

This is a team of astrophiles, writing to prose a conceiving about the lunar colony. To support the construction of a 100,000-person Moon Colony beginning in 2050, the MCM Agency must deliver approximately 100 million metric tons of materials from Earth to the Moon in a manner that balances cost, timeline, and sustainability. We evaluated three delivery scenarios: exclusive use of the Space Elevator System, traditional rocket launches, and a hybrid combination of both.



As the development of the aerospace technology, the moon colony will come true soon in the future, with a space elevator and heavy rockets. The details of the plane are attached below: Our results indicate that a rocket-only strategy leads to excessive total cost and an extended construction timeline due to limited payload capacity, despite its operational flexibility. A space-elevator-only strategy provides stable throughput and minimal environmental impact but requires a longer time to complete early-stage construction. Compared with either single-system approach, a hybrid delivery strategy more effectively balances construction speed, cost control, and environmental impact, achieving a shorter overall timeline without incurring prohibitive costs. Sensitivity analysis further shows that this hybrid approach remains robust under moderate launch failures and elevator disruptions.

Transportation strategy as the primary logistics plan for Moon Colony construction and long-term operation. When long-term cost efficiency and environmental impact are prioritized, increasing reliance on the Space Elevator System becomes a natural strategic choice. Rockets should be used during early construction to accelerate the delivery of critical infrastructure, while the Space Elevator System should progressively become the primary logistics backbone once fully operational. For long-term operation, routine supplies and water should rely mainly on the elevator system, supplemented by on-site water recycling to reduce recurrent transport demand and environmental impact.

Although future technological uncertainties remain, this strategy provides a practical, resilient, and sustainable pathway for building and maintaining the Moon Colony. We believe it offers clear guidance for policy planning and long-term implementation.

Sincerely,
Team # 2600308

Enabling humanity's sustainable expansion beyond Earth !

Report on Use of AI Tools

1. OpenAI ChatGPT (GPT-5.2, 2025 version)

Query:

What are some commonly discussed real-world challenges and considerations related to building a large-scale Moon colony?

Output

ChatGPT summarized widely cited background considerations from publicly available sources, including long-term logistics, life-support sustainability, material transportation constraints, and human health challenges in reduced-gravity environments. The response focused on qualitative descriptions and high-level context, providing background understanding without proposing any specific mathematical models, assumptions, or quantitative estimates. This information was used solely to support the introductory background discussion and to improve contextual completeness of the problem statement.

2. Anthropic Claude (Claude 3.5, latest version)

Query:

Are there common environmental concerns discussed when comparing different space transportation technologies?

Output

Claude provided a structured overview of environmental considerations frequently mentioned in literature, such as atmospheric emissions from rocket launches, energy consumption patterns, and potential advantages of electrically powered systems. The response emphasized qualitative comparisons and commonly accepted viewpoints, without suggesting metrics, optimization objectives, or evaluative frameworks. This information was used only for general background explanation and did not influence the design or results of any mathematical models.

3. GitHub Copilot (2025 version)

Query:

Can you suggest standard ways to organize Python scripts for simulations and data visualization?

Output

GitHub Copilot suggested commonly used code organization practices, including separating data loading, computation, and visualization into distinct functions, as well as adding comments and consistent naming conventions. The suggestions were limited to code structure and readability, while all algorithmic logic, numerical parameters, and simulation outcomes were independently developed and verified by the authors.

4. OpenAI ChatGPT (GPT-5.2, background reference use)

Query:

What types of water supply and recycling issues are typically discussed for long-duration space habitats?

Output

ChatGPT summarized commonly discussed water-related challenges in space habitats, including storage limitations, recycling efficiency, system reliability, and the importance of redundancy for long-term missions. The response remained descriptive and conceptual, serving only as general background knowledge. No numerical assumptions, system designs, or modeling strategies were adopted directly from this information.